

$$f : \mathbb{R}^4 \rightarrow \mathbb{R}$$

$$\begin{aligned} \max f(x) &= 5x_1 + 4x_2 + 3x_3 + 2x_4 \\ 2x_1 + x_2 + x_3 + 2x_4 &\leq 8 \\ x_1 + 3x_2 + 2x_3 + x_4 &\leq 9 \\ x_1, x_2, x_3, x_4 &\geq 0 \end{aligned}$$

Problema Duale

$$g : \mathbb{R}^2 \rightarrow \mathbb{R}$$

$$\begin{aligned} \min g(y) &= 8y_1 + 9y_2 \\ 2y_1 + y_2 &\geq 5 \\ y_1 + 3y_2 &\geq 4 \\ y_1 + 2y_2 &\geq 3 \\ 2y_1 + y_2 &\geq 2 \\ y_1, y_2 &\geq 0 \end{aligned}$$

$c^T = (8, 9, 0, 0, 0, 0) \geq 0$

Il problema è **Dualmente** ammissibile.

Metodo del Simplexso Duale

Forma Standard

$$\begin{aligned} \min g(y) &= 8y_1 + 9y_2 \\ -2y - y_2 + s_1 &= -5 \\ -y_1 - 3x_2 + s_2 &= -4 \\ -y_1 - 2y_2 + s_3 &= -3 \\ -2y_1 - y_2 + s_4 &= -2 \\ y_1, y_2, s_1, s_2, s_3, s_4 &\geq 0 \end{aligned}$$

Forma Standard

$$\begin{aligned} \min g(y) &= 8y_1 + 9y_2 \\ 2y + y_2 - s_1 &= 5 \\ y_1 + 3x_2 - s_2 &= 4 \\ y_1 + 2y_2 - s_3 &= 3 \\ 2y_1 + y_2 - s_4 &= 2 \\ y_1, y_2, s_1, s_2, s_3, s_4 &\geq 0 \end{aligned}$$

	y_1	y_2	s_1	s_2	s_3	s_4	b
s_1	-2	-1	1	0	0	0	-5
s_2	-1	-3	0	1	0	0	-4
s_3	-1	-2	0	0	1	0	-3
s_4	-2	-1	0	0	0	1	-2
	8	9	0	0	0	0	0

Entra y_1 ed esce s_1 :

	y_1	y_2	s_1	s_2	s_3	s_4	b
s_1	-2	-1	1	0	0	0	-5
s_2	-1	-3	0	1	0	0	-4
s_3	-1	-2	0	0	1	0	-3
s_4	-2	-1	0	0	0	1	-2
	8	9	0	0	0	0	0

R4 $\leftarrow$ R4 - R1
R1 $\leftarrow$ R1/2

$$\begin{aligned} R2 &\leftarrow R2 + R1 \\ R3 &\leftarrow R3 + R1 \\ R5 &\leftarrow R5 - R1 \cdot 8 \end{aligned}$$

	y_1	y_2	s_1	s_2	s_3	s_4	b
y_1	1	$\frac{1}{2}$	$-\frac{1}{2}$	0	0	0	$\frac{5}{2}$
s_2	0	$-\frac{5}{2}$	$-\frac{1}{2}$	1	0	0	$-\frac{3}{2}$
s_3	0	$-\frac{3}{2}$	$-\frac{1}{2}$	0	1	0	$-\frac{1}{2}$
s_4	0	0	-1	0	0	1	3
	0	5	4	0	0	0	-20

Entra y_2 ed esce s_2 :

	y_1	y_2	s_1	s_2	s_3	s_4	b
y_1	1	$\frac{1}{2}$	$-\frac{1}{2}$	0	0	0	$\frac{5}{2}$
s_2	0	$-\frac{5}{2}$	$-\frac{1}{2}$	1	0	0	$-\frac{3}{2}$
s_3	0	$-\frac{3}{2}$	$-\frac{1}{2}$	0	1	0	$-\frac{1}{2}$
s_4	0	0	-1	0	0	1	3
	0	5	4	0	0	0	-20

$$\begin{aligned} R1 &\leftarrow R1 + R2 \cdot \frac{1}{5} \\ R3 &\leftarrow R3 + R2 \cdot -\frac{3}{2} \\ R2 &\leftarrow R2 \cdot -\frac{2}{5} \\ R5 &\leftarrow R5 - R1 \cdot 5 \end{aligned}$$

	y_1	y_2	s_1	s_2	s_3	s_4	b
y_1	1	0	$-\frac{3}{5}$	$\frac{1}{5}$	0	0	$\frac{11}{5}$
y_2	0	1	$\frac{1}{5}$	$-\frac{2}{5}$	0	0	$\frac{3}{5}$
s_3	0	0	$-\frac{1}{5}$	$-\frac{3}{5}$	1	0	$\frac{2}{5}$
s_4	0	0	-1	0	0	1	3
	0	0	3	2	0	0	-23

$$\begin{aligned} y_1 &= \frac{11}{5} \\ y_2 &= \frac{3}{5} \\ s_1 &= 0 \\ s_2 &= 0 \\ s_3 &= \frac{2}{5} \\ s_4 &= 3 \end{aligned}$$

$$\min g(y) = 8y_1 + 9y_2 \rightarrow 8\left(\frac{11}{5}\right) + 9\left(\frac{3}{5}\right) = \frac{88 + 27}{5} = \frac{115}{5} = \boxed{23}$$

Teorema della dualità forte

$\max f(x) = \min g(y)$

$$\max f(x) = \boxed{23}$$

Scarti Complementari

Si utilizzano gli scarti complementari per ricavare la x del problema primale:

$$\boxed{\begin{array}{l} y_i[(Ax)_i - b_i] = 0 \\ [(A^T y)_i - c_i]x_i = 0 \end{array}}$$

$$\left[\frac{22}{5} + \frac{3}{5} - 5\right] x_1 = [0]x_1 = 0 \rightarrow x_1 \neq 0$$

$$\left[\frac{11}{5} + \frac{9}{5} - 4\right] x_2 = [0]x_2 = 0 \rightarrow x_2 \neq 0$$

$$\left[\frac{11}{5} + \frac{6}{5} - 3\right] x_3 = \left[\frac{2}{5}\right] x_3 = 0 \rightarrow x_3 = 0$$

$$\left[\frac{22}{5} + \frac{3}{5} - 2\right] x_4 = [3]x_4 = 0 \rightarrow x_4 = 0$$

$$\frac{11}{5}[8 - 2x_1 + x_2 + x_3 + 2x_4] = 0 \rightarrow \frac{11}{5} > 0$$

$$[8 - 2x_1 + x_2 + x_3 + 2x_4] = 0$$

$$2x_1 + x_2 + x_3 + 2x_4 = 8 \rightarrow \text{Vincolo attivo}$$

$$\frac{3}{5}[9 - x_1 + 3x_2 + 2x_3 + x_4] = 0 \rightarrow \frac{3}{5} > 0$$

$$[9 - x_1 + 3x_2 + 2x_3 + x_4] = 0$$

$$x_1 + 3x_2 + 2x_3 + x_4 = 9 \rightarrow \text{Vincolo attivo}$$

$$\begin{cases} 2x_1 + x_2 = 8 \\ x_1 + 3x_2 = 9 \end{cases}$$

$$\begin{cases} 2x_1 = 6 \\ \frac{5}{2}x_2 = 5 \end{cases}$$

$$\begin{cases} x_1 = 3 \\ x_2 = 2 \end{cases}$$

$$x_1 = 3$$

$$x_2 = 2$$

$$x_3 = 0$$

$$x_4 = 0$$

$$\max f(x) = 5x_1 + 4x_2 + 3x_3 + 2x_4 \rightarrow 5(3) + 4(2) + 3(0) + 2(0) = 15 + 8 = \boxed{23}$$